

Parallel to yz plane then the volume is given by

$$S(x) dx$$

For given x ,

$$S(x) = \int_{y_{\min}(x)}^{y_{\max}(x)} f(x, y) dy$$
$$\iint_R f(x, y) dA = \int_{x_{\min}}^{x_{\max}} \left(\int_{y_{\min}(x)}^{y_{\max}(x)} f(x, y) dy \right) dx$$

Iterated Integral

Example

$$z = 1 - x^2 - y^2$$

Region:

$$0 \leq x \leq 1$$

$$0 \leq y \leq 1$$

$$\int_0^1 \int_0^1 (1 - x^2 - y^2) dy dx$$

1) Inner integral

$$\int_0^1 (1 - x^2 - y^2) dy = \left[y - x^2 y - \frac{y^3}{3} \right]_0^1$$
$$= 1 - x^2 - \frac{1}{3}$$
$$= \frac{2}{3} - x^2$$

Outer integral

$$\int_0^1 \left(\frac{2}{3} - x^2 \right) dx$$
$$= \left[\frac{2x}{3} - \frac{x^3}{3} \right]_0^1$$
$$= \frac{1}{3}$$

$$dA \Rightarrow dy \, dx$$

Because for a small rectangle,

$$\Delta A = \Delta x \Delta y$$

Example 2

R = quarter disk

$$x^2 + y^2 \leq 1$$

$$x \geq 0, \quad y \geq 0$$

For given x , the range of y is from

$$y = 0 \quad \text{to} \quad y = \sqrt{1 - x^2}$$

$$x^2 + y^2 = 1$$

$$y = \sqrt{1 - x^2}$$

$$\iint_R (1 - x^2 - y^2) \, dA = \int_0^1 \int_0^{\sqrt{1-x^2}} (1 - x^2 - y^2) \, dy \, dx$$

Inner integral

$$\begin{aligned} \int_0^{\sqrt{1-x^2}} (1 - x^2 - y^2) \, dy &= \left[y - x^2 y - \frac{y^3}{3} \right]_0^{\sqrt{1-x^2}} \\ &= \sqrt{1-x^2} - x^2 \sqrt{1-x^2} - \frac{(1-x^2)^{3/2}}{3} \\ &= \frac{2}{3} (1-x^2)^{3/2} \end{aligned}$$

Outer integral

$$\int_0^1 \frac{2}{3} (1-x^2)^{3/2} \, dx$$

Let

$$x = \sin \theta$$

Then

$$(1 - x^2)^{1/2} = \cos \theta$$

and

$$dx = \cos \theta \, d\theta$$

Hence,

$$\begin{aligned} \int_0^1 \frac{2}{3} (1 - x^2)^{3/2} dx &= \int_0^{\pi/2} \frac{2}{3} \cos^3 \theta \cdot \cos \theta \, d\theta \\ &= \frac{2}{3} \int_0^{\pi/2} \cos^4 \theta \, d\theta \end{aligned}$$

Using

$$\cos^2 \theta = \frac{1 + \cos 2\theta}{2}$$

we get

$$= \frac{2}{3} \int_0^{\pi/2} \left(\frac{1 + \cos 2\theta}{2} \right)^2 d\theta$$

Example 1

$$\int_0^1 \int_0^2 dy \, dx$$

Example 2

$$\begin{aligned} \int_0^1 \int_0^{\sqrt{x}} \frac{e^y}{y} dy \, dx \\ = \int_0^1 \int_{y^2}^y \frac{e^y}{y} dx \, dy \end{aligned}$$

Inner

$$\left[\frac{e^y}{y} x \right]_{x=y^2}^y = e^y - ye^y$$

Outer

$$\int_0^1 (e^y - ye^y) dy = [ye^y + 2e^y]_0^1 = e - 2$$

Lecture 7: Double Integrals in Polar Coordinates Application

$$\iint_R f(x, y) \, dA$$

$$= \int \int f(x, y) dy dx$$

$$\iint_{x^2+y^2 \leq 1, y \geq 0} (1 - x^2 - y^2) dA$$

In polar coordinates,

$$x = r \cos \theta$$

$$y = r \sin \theta$$

$$dA = r dr d\theta$$

$$\int_0^{\pi/2} \int_0^1 f r dr d\theta$$

$$f = 1 - x^2 - y^2$$

$$= 1 - (x^2 + y^2)$$

$$= 1 - r^2$$

$$\int_0^{\pi/2} \int_0^1 (1 - r^2) r dr d\theta$$

$$= \int_0^{\pi/2} \left[\frac{r^2}{2} - \frac{r^4}{4} \right]_0^1 d\theta$$

$$= \int_0^{\pi/2} \frac{1}{4} d\theta = \frac{1}{4} \cdot \frac{\pi}{2} = \frac{\pi}{8}$$

Applications

Find area of region R

$$\text{Area}(R) = \iint_R 1 dA$$

Mass of a flat object with density δ = mass per unit area

$$\Delta m = \delta \Delta A$$

$$\text{mass} = \iint_R \delta dA$$

1. Average value in R

Average of f :

$$\bar{f} = \frac{1}{\text{Area}(R)} \iint_R f \, dA$$

Weighted average of f with density δ :

$$\frac{1}{\text{mass}(R)} \iint_R f \delta \, dA$$

2. Center of mass of a plane object with density (δ)

$$(\bar{x}, \bar{y})$$

$$\bar{x} = \frac{1}{\text{mass}} \iint x \delta \, dA, \quad \bar{y} = \frac{1}{\text{mass}} \iint y \delta \, dA$$

3. Moment of inertia

Mass = how hard it is to impart translational motion.

Moment of inertia = rotational motion about an axis.

Idea: Kinetic energy

$$\frac{1}{2}mv^2$$

for mass m at distance r and angular velocity ω .